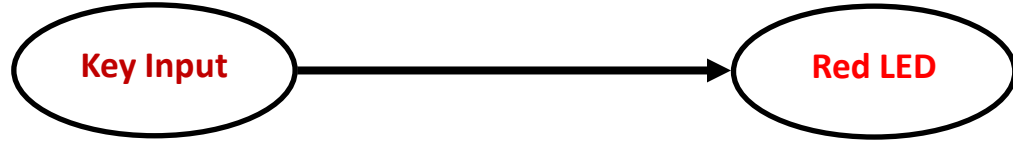
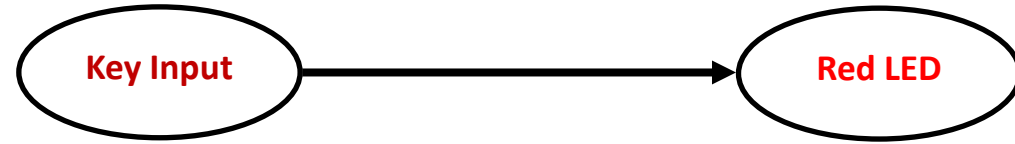


Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads

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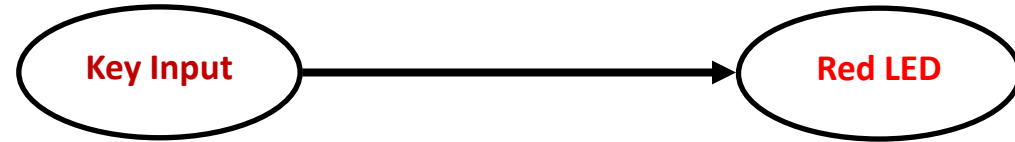


Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads



Thread
Node
Process
Cell
Task
Function

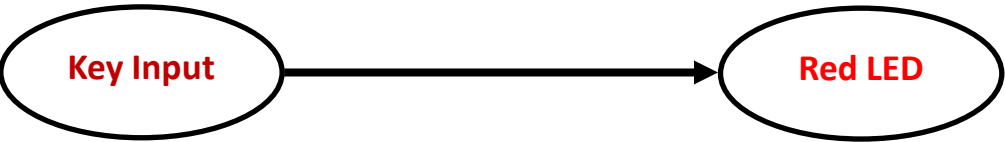
Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads



Thread
Node
Process
Cell
Task
Function

Channel
Topic
Socket
Queue
FIFO
Message

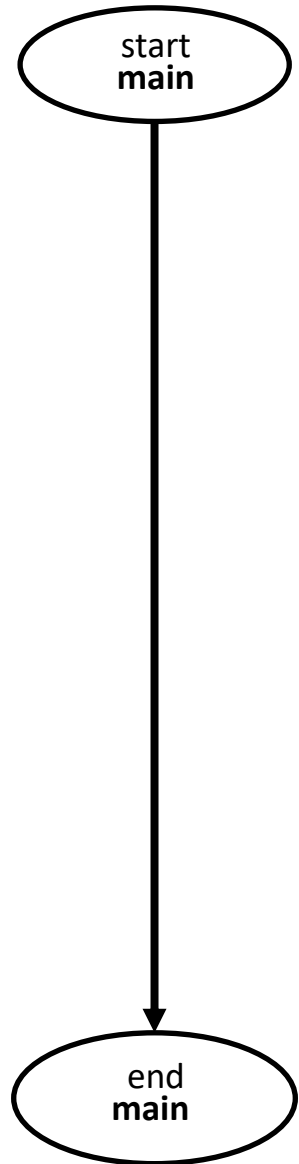
Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads



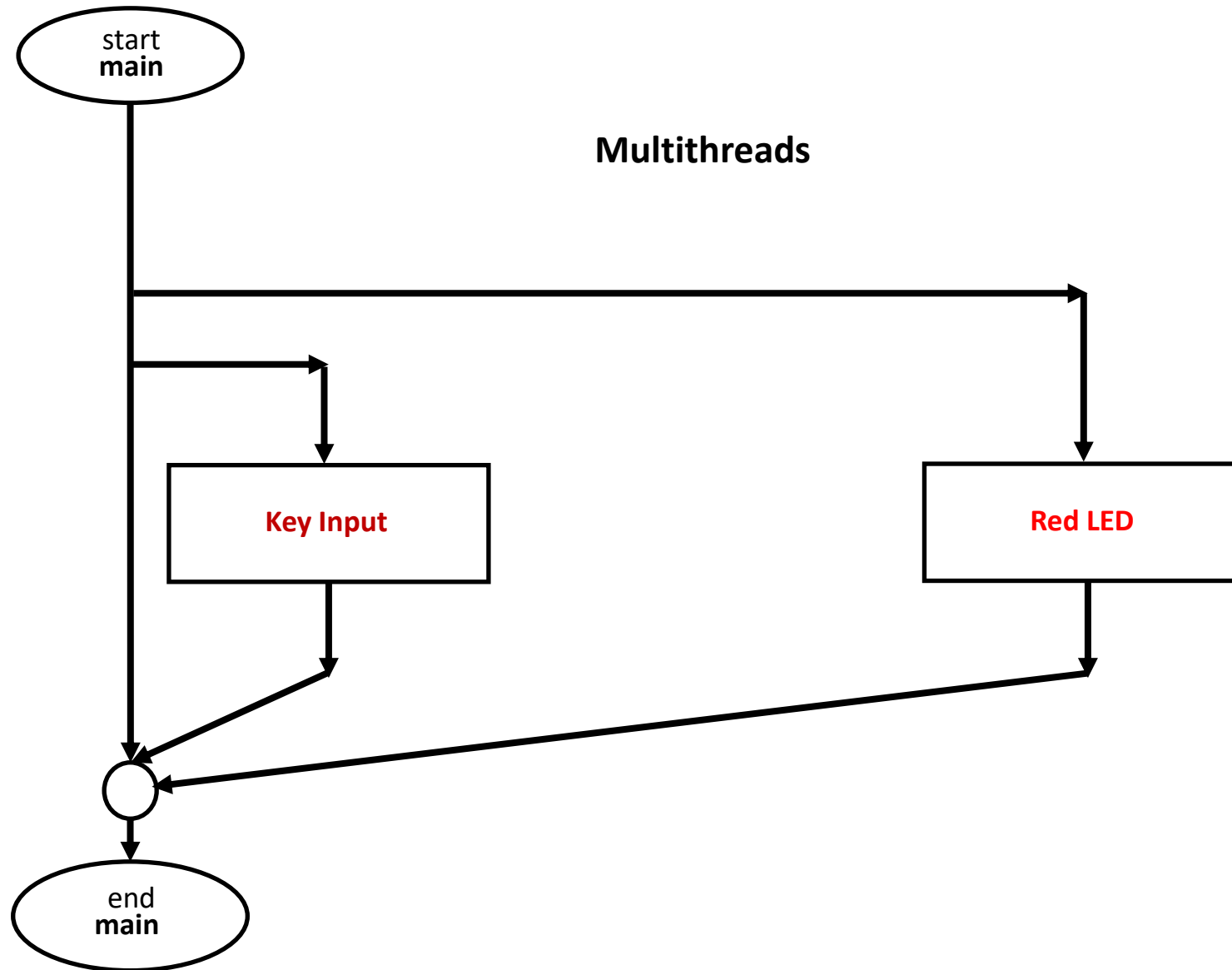
Thread	Channel
Node	Topic
Process	Socket
Cell	Queue
Task	FIFO
Function	Message

- in single processor
 - in multiple cores
 - in multiple computers
 - in distributed computers
- PAN: Personal Area Network
LAN: Local Area Network
MAN: Metropolitan Area Network
WAN: Wide Area Network
- Wired Network
Wifi
Cellular Network, Mobile Network, 5G

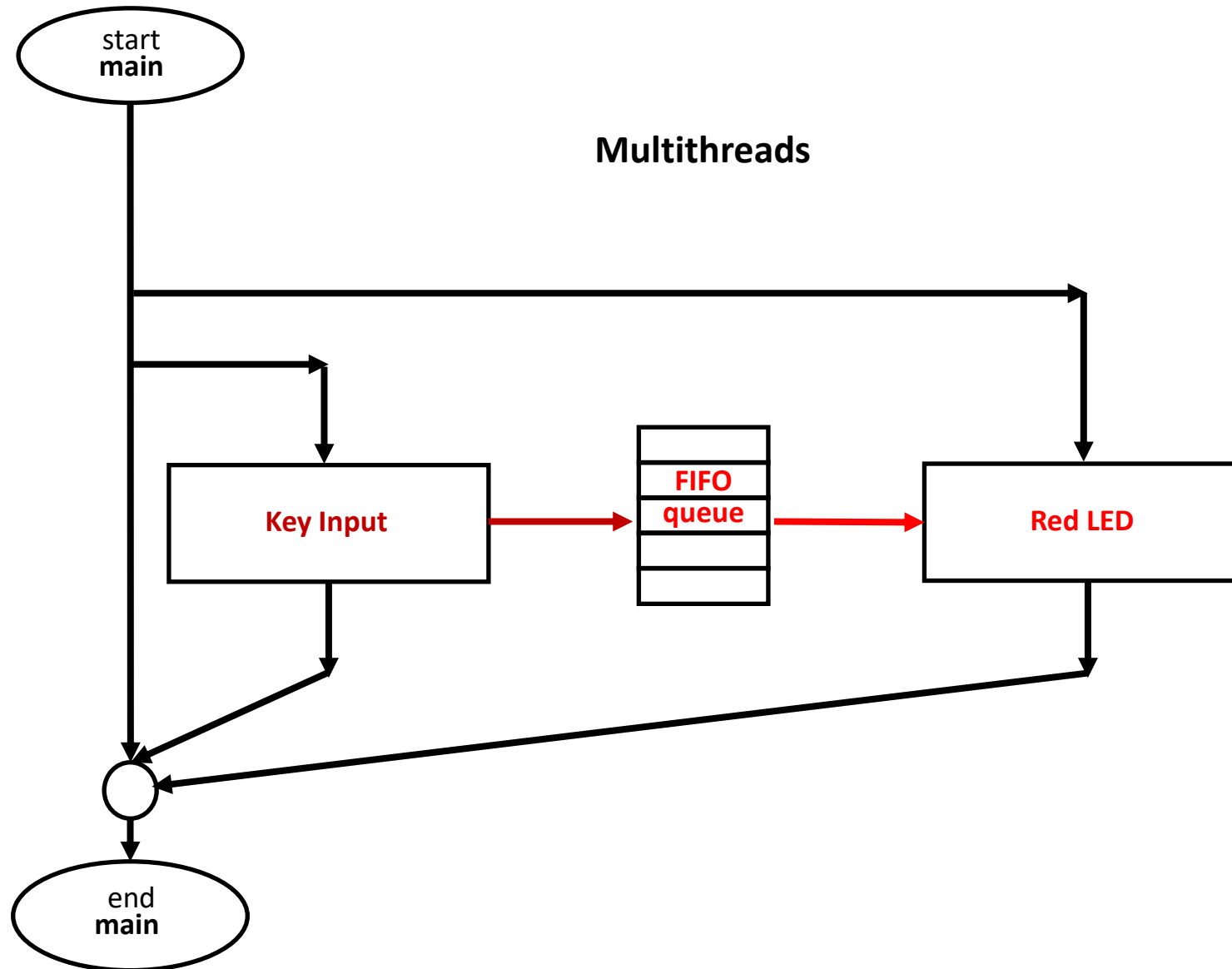
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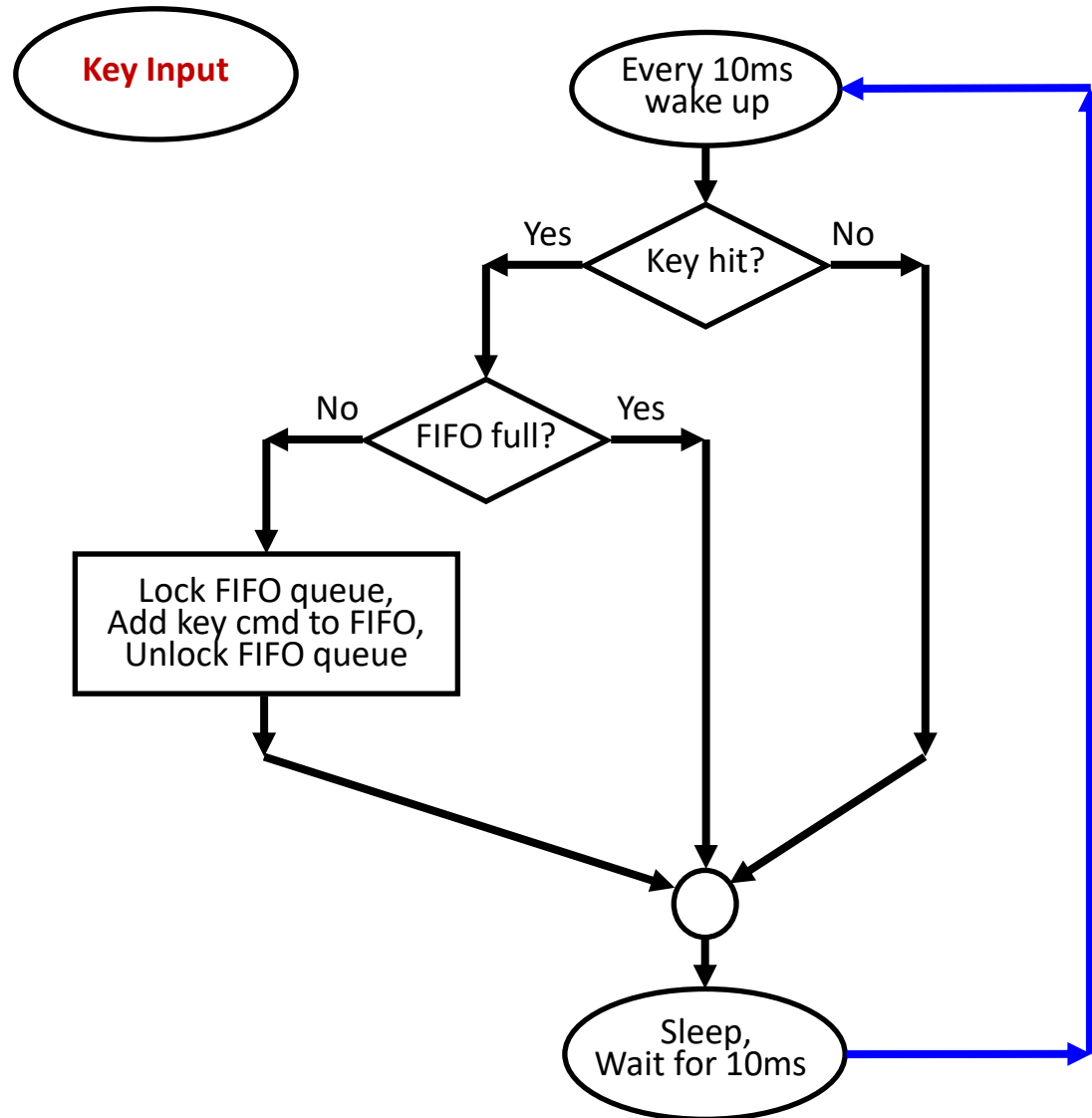


Key Input

Simple Key Input function threads example program, with FIFO queues for command

- * **Key Input** function thread and **Red** LED on/off driver function thread works together
- * Simple Key Input function thread wakes up every **10ms** and does the following:
 - 1 **Read** key: if no key hit, then skip to step 3 below
 - 2 **Place** key character on the FIFO queue, if the queue not full
 - 3 **Do-nothing**, to wait 10ms
- * Simple Key Input function reads keyboard and place the character on the **FIFO queue**, length = 1024
- * Only one keyboard reading made in 10ms duration
- * Available Key commands are as follows: (other keys are ignored)
 - 'n' – **ON** command, turn on LED
 - 'f' – **OFF** command, turn off LED
 - 'q' – **Quit** command, quit this Key Input function thread and LED driver function thread
- * Adding CMD to **FIFO queue**: always check the FIFO queue FULL before adding more data, always lock first (mutex) the FIFO queue, then add data to the FIFO queue, and unlock the FIFO queue

Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads



Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads

A red oval containing the text "Red LED" in red font.

Simple LED on/off driver function threads example program, with FIFO queues for command

- * **Red** LED on/off driver function thread and **Key Input** function thread works together
- * Simple LED on/off driver function thread wakes up every **10ms** and does the following:
 - 1 **Check** FIFO queue empty: if EMPTY, then skip to step 3 below
 - 2 **Take** CMD key character from the FIFO queue, and do the CMD: turn ON/OFF Red LED
 - 3 **Do-nothing**, to wait 10ms
- * Simple LED on/off driver function takes CMD character from the **FIFO queue**, length = 1024
- * Only one CMD executed in 10ms duration
- * Available Key commands are as follows: (other keys are ignored)
 - 'n' – **ON** command, turn on LED
 - 'f' – **OFF** command, turn off LED
 - 'q' – **Quit** command, quit this LED on/off driver function thread
- * Taking CMD from the **FIFO queue**: always check the FIFO queue EMPTY before taking data, always lock first (mutex) the FIFO queue, then take data from the FIFO queue, and unlock the FIFO queue

Programming Multithreads and Message FIFO Queue Example: hw0c6fifo1simp.c - Simple Message Producer and Message Receiver Threads

